Assignment 2: Diabetes Classification

Understanding and evaluating a classification workflow

Group size: approximately 3 students

1. Purpose

Use the supplied diabetes_binary_classification_data.csv to practise an end-to-end classification workflow. The goal is not to obtain the highest possible accuracy. The focus is on understanding the data, using EDA to answer useful questions, keeping TEST protected, fitting Logistic Regression correctly, and interpreting classification predictions and errors.

UNDERSTAND -> EDA -> SPLIT -> PREPARE -> FIT -> PREDICT -> EVALUATE -> INTERPRET

2. Group work

Complete the assignment in groups of approximately 3 students. You may divide implementation tasks, but every group member must understand the complete solution.

During the feedback discussion, any group member may be asked to explain an important part of the workflow, including a part implemented by another member.

3. Files

- diabetes_binary_classification_data.csv
- Diabetes Classification - Understanding the Dataset

Use the Understanding the Dataset guide when you need the meaning, values, or variable type of a feature. You are not expected to know the health-variable codes from memory.

4. What your notebook should show

Include code, useful outputs and short explanations. For the important parts of your analysis, think in terms of:

QUESTION -> EVIDENCE -> DECISION

Step 1 - Understand the data

- Identify the input features X and the target y.
- Explain what class 0 and class 1 mean.
- Check the number and proportion of observations in each class.
- Choose three features of different types and briefly explain what their values mean. Use the Understanding the Dataset guide.

Step 2 - Explore the data

Use EDA to investigate all input features and their relationships with Diabetes_binary. Choose plots or tables that are appropriate for each type of feature:

- for numerical features, examine their distributions and compare them between the two target classes;
- for binary categorical features, examine counts/proportions and compare the proportion of Diabetes_binary across the two categories;
- for ordered categorical features, examine how the distribution or proportion of Diabetes_binary changes across the ordered categories.

Explore all input features, but focus your detailed interpretation on the most informative or interesting results. You may use loops or functions where appropriate rather than repeating essentially identical code. For the results you discuss in detail, state the question, show appropriate evidence, explain what you observe, and explain what the evidence allows

you to conclude and what it does not. Look for patterns such as differences between the two target classes, changes in the proportion of class 1 across categories, unusual distributions, and features that appear to have little relationship with the target. Remember: EDA describes associations and patterns in the observed data; an association does not show that one variable causes diabetes.

Step 3 - Create TRAIN and TEST

- Split X and y into TRAIN and TEST.
- Use stratify=y so that TRAIN and TEST have approximately the same proportion of class 0 and class 1.
- Keep TEST separate until evaluation. Do not balance or resample TEST.

Step 4 - Prepare the features

Prepare the input features using the information in the Understanding the Dataset guide and the preprocessing methods introduced in the course. Your preprocessing choice should reflect whether a feature is numerical, binary categorical, or ordered categorical. Briefly justify how you represent each type. There are no missing values in this dataset, so missing-value imputation is not required.

- Use a Pipeline/ColumnTransformer where appropriate.
- Anything learned from the data during preprocessing must be learned from TRAIN only and then applied to TEST.

Step 5 - Train Logistic Regression

Use Logistic Regression, as introduced in the classification lecture. Fit the complete Pipeline using TRAIN.

- Use model.predict(X_test) to obtain predicted classes.
- Use model.predict_proba(X_test)[:, 1] to obtain predicted probabilities for class 1.

Step 6 - Evaluate the model

Evaluate the Logistic Regression model on TEST using:

- confusion matrix;
- accuracy;
- precision;
- recall;
- F1 score.

Explain what a false positive and a false negative mean for this diabetes classification problem. Do not judge the model from accuracy alone.

Step 7 - Probability and threshold

Use the predicted probabilities from Logistic Regression to compare two decision thresholds:

- Threshold 0.50
- Threshold 0.40

For each threshold, calculate the confusion matrix, precision and recall. Compare the results. Explain what happened to false positives, false negatives, precision and recall. Also explain what did NOT change when you changed only the threshold.

Step 8 - Conclusion

- Summarize the most important things you learned from the EDA and the Logistic Regression results.
- Give at least two limitations of this analysis.
- Do not interpret an association as evidence that one feature causes diabetes.

5. Questions you must answer

Q1. Understanding the data

What do class 0 and class 1 mean? Is the target balanced? Why does the class distribution matter when evaluating the model?

Q2. Understanding the features

Choose three features of different types. What do their values mean? Why is looking only at the numbers stored in the CSV not enough to understand a feature?

Q3. EDA

You explored all input features. Which three to five findings were most informative or interesting? What did you observe? What can you conclude from these results, and what can you not conclude?

Q4. TRAIN and TEST

Why must TEST remain separate while preprocessing and the model are fitted?

Q5. Classification errors and metrics

Using your confusion matrix, explain false positives and false negatives for this diabetes problem. What do accuracy, precision and recall tell you? Why is accuracy alone not enough?

Q6. Probability and threshold

Compare thresholds 0.50 and 0.40. What happened to false positives, false negatives, precision and recall? What did not change when you changed only the threshold?

Q7. Conclusion

What did you learn from the data and the Logistic Regression model? Give at least two limitations of this analysis.

6. Threshold comparison table

Threshold	Accuracy	Precision	Recall	F1	False positives	False negatives
0.50						
0.40						

7. Focus of model evaluation

Calculate and report the required evaluation metrics for your model and for the two decision thresholds. There is no required minimum value that your model must achieve. Your task is to correctly interpret the results and explain the trade-offs between the metrics, rather than optimize the model to reach a particular performance value.

8. Submission

- GitHub repository containing the completed notebook, code, relevant outputs and visualizations.
- Short written document answering Q1-Q7 and the two group-work questions G1-G2. Refer to evidence from your notebook; do not copy all code into the written answers.
- Brief group-work reflection. Answer these two questions:
- G1. How was the work divided among group members, and what did each member contribute?
- G2. How did your group make sure that everyone understood the complete solution, not only the part they worked on?

9. Feedback discussion

Any group member may be asked to explain a feature meaning, an EDA result, the TRAIN/TEST split, a preprocessing step, a confusion-matrix entry, precision versus recall, or what changed when the threshold changed. The focus is understanding and reasoning, not memorizing code.

Meaning first -> evidence -> decision. Keep TEST protected. Association is not causation. Accuracy is not enough by itself.